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Role of neural oscillations in depression: highlights on gamma oscillations

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Abstract: Depression is a prevalent and debilitating psychiatric disorder traditionally conceptualized as monoaminergic dysregulation. Although this view has guided pharmacotherapeutic development for decades, its clinical and mechanistic limitations have spurred research into alternative frameworks, particularly those focusing on network-level dysfunctions. Within this evolving framework, neural oscillations, especially in the frequency of gamma oscillations (30-80 Hz), serve as a critical window into the dynamics of the large-scale brain network. Therefore, this review summarizes a substantial body of evidence positing a central role for aberrant gamma oscillations in the pathophysiology of depression. We first synthesize the neurobiological basis of gamma oscillations and critique their crucial relationship with synaptic plasticity, proposing that gamma impairments may underpin key cognitive and affective deficits observed in this disorder. Notably, we critically examine the complex effects of conventional antidepressants, including selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs), and neuromodulation therapies like repetitive transcranial magnetic stimulation (rTMS) and deep brain stimulation (DBS) on

gamma oscillations. Finally, we discuss emerging gamma oscillation-targeting therapies like 40 Hz flickering light therapy and future directions. In summary, we conclude that gamma oscillations represent not an epiphenomenon but a fundamental pathophysiological hub and a promising therapeutic target, paving the way for more precise and effective treatments for depression.

Keywords: Gamma oscillations; Depression; SSRIs; Antidepressant; Ketamine

1 Introduction

Depression is a prevalent and debilitating psychiatric disorder and currently affects 350 million people worldwide. Depression has traditionally been conceptualized as monoaminergic dysregulation, positing deficits in brain monoamine levels (Berton & Nestler, 2006; Pittenger & Duman, 2008). This hypothesis has guided the development of first-line antidepressants for decades, such as SSRIs and SNRIs. However, these antidepressants face considerable limitations, including delayed onset of action, low response rates (50-60%), and side effects. Beyond this chemical imbalance, some studies indicate that information-processing dysfunction within neural networks might underlie depression (Castrén, 2005; Leistedt & Linkowski, 2013), highlighting the need to look beyond neurotransmitter deficiencies and examine dysfunctions in large-scale brain networks. Furthermore, the clinical diagnosis of depression mainly relies on subjective clinical symptom scales and doctor interviews, lacking objective biological indicators and sensitive diagnostic methods in clinical practice, which poses a significant challenge for

evaluating the therapeutic effects of antidepressant interventions.

Neural oscillations, an important form of brain networks, originate from the coordination between the local neural dynamics and the large-scale structure of the brain, which have been reported to be involved in neuropsychiatric disease (Başar, 2013). This large-scale coordination is fundamentally enabled by neural oscillations (Buzsáki & Draguhn, 2004) and allows different parts of the brain to communicate and orchestrate responses to internal and external stimuli, which is essential for a healthy brain. Increasing evidence indicates that dysregulation of these oscillatory rhythms represents a core pathophysiological feature of several neuropsychiatric disorders (Başar, 2013). Within this framework, gamma oscillations (30-80 Hz), an important band of neural oscillations, have garnered attention for their role in depression.

Electrophysiological studies in humans and animal models collectively suggest that depression is associated with significant abnormalities in gamma oscillations (Li et al., 2015; Sauer et al., 2015; Strelets et al., 2007; Yin et al., 2023; Yin et al., 2024). In addition, gamma oscillations are implicated in various antidepressant treatments, including conventional medications such as SSRIs and SNRIs (Akhmetshina et al., 2016; Alaiyed et al., 2019; Papp et al., 2018) and neuromodulation therapies like rTMS and DBS (Bailey et al., 2018; Pathak et al., 2016; Scangos et al., 2021; Sun et al., 2015). Beyond these, ketamine, exerting rapid-onset antidepressant actions in clinical practices (Abbar et al., 2022; Zolghadriha et

al., 2024), has been reported to regulate gamma oscillations (Kantor et al., 2023; Yin et al., 2024). Collectively, these findings underscore the value of reviewing recent advances in gamma oscillation research to guide the development of next-generation antidepressants targeting brain networks (**Graphical Abstract**).

2 Methods

To summarize the research progress of neural oscillations in depression, we focus our review on gamma oscillations. Literatures were obtained on PubMed from clinical and animal studies published in English between 2000 and 202. Search terms included “Neural oscillations” or “Gamma oscillations” or “ γ oscillations” AND “Chronic stress” or “Depression” or “Synaptic plasticity” or “Antidepressants” or “SSRIs” or “SNRIs” or “TCAs” or “Ketamine” or “Neuromodulation therapies” or “DBS” or “rTMS” or “Flickering light therapy” or “Dopamine” or “Serotonin”. It should be noted that literature unavailable in full text through standard academic channels was excluded.

3 Neural oscillations

Neural oscillations, as a fundamental phenomenon in neuroscience, are rhythmic patterns of neural activities that represent the synchronization of local neuron populations or neuron assemblies in the central nervous system (Buzsáki & Draguhn, 2004). Neural oscillations are proposed to provide a temporal scaffold for coordination of neuronal network activity and routing of information (Li et al., 2023; Palva & Palva, 2018), and could be detected by electroencephalography (EEG), local

field potentials (LFPs), electrocorticography (ECoG), and magnetoencephalography (MEG) in animal models and clinical settings. Neural oscillations govern the timing of neuronal spikes at the microscale, and at the macroscale, they coordinate the dispersed cortical communication to enable temporal and spatial brain connectivity (Lowet et al., 2015; Zhang et al., 2018). Neural oscillations are classified based on frequency into the delta (0.1-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-30 Hz), and gamma (30-80 Hz) oscillations (**Figure 1**) (Cole & Voytek, 2017). Notably, the translational value of gamma oscillations in antidepressant treatments, such as their potential clinical relevance, is currently receiving attention among these neural oscillations. Therefore, this review aims to summarize key findings regarding gamma oscillations as a biomarker or endophenotype for depression, and provides a detailed analysis of these issues, with an eye toward the future of this emerging topic.

4 Neurobiological basis of gamma oscillations

Gamma oscillations are a prominent phenomenon in the central nervous system (CNS) that coordinate connectivity between brain regions, underpinning critical functions such as memory formation, sensory processing, and mood regulation (Krupa et al., 2014). Due to the scope of this review, we focused on the role of gamma oscillations in neuropsychiatric disorders, particularly depression. Gamma oscillations are observed in multiple brain regions implicated in depression, including the medial prefrontal cortex (mPFC) (Yin et al., 2023; Yin et al., 2024), the hippocampus (Krause & Jia, 2005; Wang et al., 2022), and the nucleus

accumbens (NAc) (Akhoondian et al., 2023; Goda et al., 2013). At the cellular level, gamma oscillations emerge from the precise interaction between excitatory glutamatergic neurons and inhibitory GABAergic interneurons (Li et al., 2021; Rhodes et al., 2025). This interplay maintains a functional excitatory-inhibitory (E-I) balance, a mechanism crucial for neural circuit operation and deeply implicated in the pathophysiology of depression and the action of antidepressant treatments (Page & Coutellier, 2019; Wang et al., 2019; Yin et al., 2021). Key cellular orchestrators of these rhythms are fast-spiking, including parvalbumin-expressing (PV+) and somatostatin-expressing (SST+) interneurons, which have a key role in the generation of gamma oscillations (Antonoudiou et al., 2020; Onorato et al., 2025). Notably, these two inhibitory interneurons both play an important role in the pathophysiology of depression (Asim et al., 2024; Fogaça et al., 2021; Zhang et al., 2017). In summary, the neurobiological basis of gamma oscillation generation hinges on a core E-I circuit mechanism, primarily driven by PV+ and SST+ interneurons. These gamma oscillation activities are not only fundamental to inter-regional communication but also originate within a limbic network central to mood regulation

5 Gamma oscillation abnormalities in depression

Despite their established role in depression (Fitzgerald & Watson, 2018), gamma oscillations show inconsistent directional alterations across studies. This section critically reviews these findings to examine how methodological and biological

factors contribute to the observed variability and underscore the central role of gamma oscillations in depression pathophysiology.

5.1 Preclinical evidence

Preclinical studies across diverse animal models consistently report alterations in gamma oscillations associated with depression-like states, but the direction of change can vary by brain region and model type (**Table 1**). Preclinical evidence for gamma oscillation alterations in depression-related states can be broadly categorized into two lines of investigation. The first line of evidence derives from genetic or predisposed animal models exhibiting inherent depression-like phenotypes. For example, two groups respectively reported that Flinders sensitive line (FSL) rats and mice expressing the truncated Disrupted-in-schizophrenia 1 (Disc1) mutation, both displaying depression-like behavioral phenotypes, showed significant deficits in gamma oscillations (Sauer et al., 2015; Voget et al., 2015). Another line of evidence mainly comes from animal models of depression. For instance, we recently demonstrated that chronic restraint stress (CRS) for 4 weeks not only induced significant depression-like behaviors in mice, but also decreased the firing of pyramidal neurons and suppressed gamma oscillations in the mPFC (Yin et al., 2023). In addition, the decreased gamma oscillations in the mPFC of CRS-treated mice were observed in our other study (Yin et al., 2024). Consistent with findings in the CRS model, **Zhou's team** reported that chronic unpredictable mild stress (CUMS) selectively decreased high gamma oscillation power in the dorsal DG of the

hippocampus without affecting low gamma oscillation power (Zhou et al., 2022). Furthermore, *Li et al.* reported that bilateral olfactory bulbectomy (OBx), a frequently used animal model of depression (Coppola, 2025), dramatically reduced gamma oscillations in the piriform cortex (PirC) and resulted in significant depression-like behaviors (Li et al., 2023). Notably, the effects of chronic stress on gamma oscillations are brain region-specific. In contrast to the suppressive effects observed in the mPFC and hippocampus, *Akhoondian and colleagues* reported that 3-week chronic mild stress (CMS) enhanced almost all band powers in the NAc, including the gamma band, accompanied by depression-like behaviors in Wistar rats (Akhoondian et al., 2023), highlighting the complexity and anatomical specificity of gamma oscillation alterations in depression models.

In addition to stress and genetic models, inflammation-related mechanisms have been linked to gamma oscillation deficits. For instance, in a recent study, we found that lipopolysaccharide (LPS) injection, a commonly used inflammation-induced model of depression, remarkably decreased gamma oscillations in the mPFC, accompanied by significant depression-like behaviors (Yin et al., 2024). Similarly, *Liu et al.* found that mice exposed to combined LPS and chronic unpredictable stress (CUS) displayed significantly decreased PV expression and disturbance of gamma oscillations in the PFC (Liu et al., 2023), and *Gao's team* reported that repeated LPS injections induced dysfunction of PV⁺ interneurons and decreased gamma oscillations in the hippocampus (Gao et al., 2023). These findings collectively

suggest that PV+ interneuron dysfunction might underlie the LPS-induced deficits in gamma oscillations.

5.2 Clinical evidence

Clinical studies indicate that gamma oscillations are significantly altered in individuals with depression (**Table 2**). However, the nature of these alterations is not uniform, with studies reporting both increases and decreases in gamma oscillations. Several studies have reported hyper-synchronization of gamma oscillations in depression. For instance, abnormally increased gamma oscillation in patients with depression has been reported during resting state EEG (Strelets et al., 2007). Building on this work, *Strelets et al.* reported that gamma oscillations increased in frontal and temporal regions in subjects with depression when performing spatial and arithmetic tasks (Strelets et al., 2007). *Li's group* reported that the global EEG coherence in gamma bands of depressed patients was significantly higher than controls, especially in the high gamma band during emotional face processing (Li et al., 2015). *Akar et al.* performed a nonlinear analysis of EEG in depression with fractal dimensions and demonstrated that individuals with depression exhibited increased resting complexity of gamma signaling in the frontal and parietal cortex (Akar et al., 2015). Supporting this notion, *Strelets et al.* performed the quantitative EEG (qEEG) analysis on suicidal ideation and suicide attempt in depression, and found that high-gamma oscillations were significantly higher in patients with suicidal ideation than in controls (Arikan et al., 2019), highlighting the significance

of gamma oscillations as a biological marker for suicide attempt in patients with depression. Recently, **Dai's group** recorded the magnetoencephalography data while participants performed a facial recognition task, and found that gamma oscillations were significantly increased in patients with major depressive disorder (MDD) (Dai et al., 2022). In addition, the latest evidence from **Tsai's team** suggested that gamma oscillations in treatment-resistant depression (TRD) were positively correlated with the symptom severity on the HDRS-17 (Tsai et al., 2025), suggesting their potential as an electrophysiological marker for the severity of TRD.

In contrast, other lines of evidence point to decreased gamma oscillatory activity in depression. For instance, MEG studies from **Lee's** and **Liu's teams** revealed that patients with MDD exhibited decreased gamma activity in the frontal cortex while performing emotion-related tasks (Lee et al., 2010; Liu et al., 2014). In addition, **Liu et al.** reported that patients with MDD showed significantly reduced gamma oscillation powers in the left temporal and the bilateral occipital regions, along with weakened gamma-band connectivity between the left hemisphere and the right frontal area (Liu et al., 2022). Similarly, in a recent study where participants performed a sad expression recognition task during MEG recording, **Dai and colleagues** demonstrated that individuals with MDD and suicide attempt (SA) exhibited attenuated activation in low gamma oscillations and decreased intra-network connectivity within the fronto-parietal region (Dai et al., 2023).

In summary, these findings point to a pattern of impaired generation and

maintenance of gamma oscillations across multiple brain regions in MDD, offering new perspectives on the neuropathological mechanisms associated with gamma dysregulation in depression. Although gamma oscillations are highly influenced by brain region, cognitive state, and clinical subtype, they hold promise as electrophysiological biomarkers for diagnosing MDD, evaluating symptom severity.

6 Gamma oscillations and synaptic plasticity

Synaptic plasticity, conferring environmental adaptability through modification of the connectivity between neurons and neuronal circuits, plays an important role in depression (Duman et al., 2016; Liu et al., 2017; Marsden, 2013; Vose & Stanton, 2017). For example, depression-like behaviors were associated with impairments of synaptic plasticity, such as neuronal atrophy and synaptic loss in the mPFC, the basolateral amygdala (BLA), and the hippocampus (Colyn et al., 2019; Ge et al., 2024). Multiple studies have shown a close relationship between synaptic plasticity and gamma oscillations (Bibbig et al., 2001). For instance, **Bibbig and colleagues** refined a computational model and replicated the gamma to beta transition *in vitro*, reported that one function of gamma oscillations is to permit synaptic plasticity (Bibbig et al., 2001), providing the initial theoretical basis for the involvement of gamma oscillations in regulating synaptic plasticity. Notably, subsequent experimental studies have strengthened this link. For example, **Xu et al.** demonstrated that impairment of cognitive function and synaptic plasticity was associated with alteration of information flow in gamma oscillations in melamine-

treated rats (Xu et al., 2013). Similarly, **Zheng & Zhang** reported that the impaired synaptic plasticity in the vCA1-mPFC pathway could be reflected by the attenuated theta-gamma cross-frequency coupling of LFPs in the depression state (Zheng & Zhang, 2015). In addition, **Zarnadze and colleagues** combined the LFP recordings *in vivo* in awake behaving mice with whole-cell recordings *in vitro* in slice preparations, and proposed that gamma oscillations represent a network state that introduced long-lasting synaptic plasticity in a cell-specific manner (Zarnadze et al., 2016). Importantly, the interdependence between neuronal oscillations and long-term potentiation (LTP), a long-term form of synaptic plasticity (He et al., 2016), has been consistently observed in preclinical studies. For example, **Kalweit et al.** have observed that a significant increase in the gamma power shortly after high-frequency stimulation (HFS) to induce LTP in the dentate gyrus of freely moving animals (Kalweit et al., 2015). Consistent with this, **Bikbaev and Manahan-Vaughan** showed that specific changes in gamma oscillations could predict the success of afferent stimulation in inducing robust LTP (Bikbaev & Manahan-Vaughan, 2017), highlighting gamma oscillations as a potential biomarker for plasticity induction. Taken together, these findings indicated that synaptic plasticity may be an important neurobiological basis for gamma oscillations regulating depression.

7 Gamma oscillations and 5-HT system

Serotonin (5-HT) is a powerful neuromodulator of neuronal activity within the central nervous system, and dysfunction in the 5-HT system has been implicated in

depression (Dell'Osso et al., 2016; Kraus et al., 2017; Mahar et al., 2014). Empirical studies suggest that 5-HT plays an important role in regulating gamma oscillations (Krause & Jia, 2005; Puig et al., 2010; Wójtowicz et al., 2009) (**Table 3**). Therefore, this section summarizes the link between gamma oscillations and the 5-HT system, focusing mainly on the 5-HT neurotransmitter and its receptor subtypes.

7.1 5-HT neurotransmitter

A substantial body of evidence from many studies suggested that 5-HT exerted an inhibitory effect on gamma oscillations. For example, an earlier *in vitro* study demonstrated that 5-HT decreases carbachol-induced power of hippocampal gamma oscillations in a concentration-dependent and reversible manner (Krause & Jia, 2005). Similarly, **Wójtowicz's group** reported that 5-HT suppressed both kainate- and carbachol-induced gamma oscillations in rat hippocampus *in vitro*, also in a dose-dependent and reversible manner (Wójtowicz et al., 2009). Subsequent *in vivo* studies further support 5-HT's inhibitory influence on gamma oscillations. For example, **Puig and colleagues** found that electrical stimulation of the dorsal raphe nucleus (DRN) in rats led to 5-HT release, which modulated prefrontal gamma oscillations and resulted in an overall reduction in gamma amplitude (Puig et al., 2010). Consistent with this suppressive effect, **Johnston et al.** used the whole-cell patch clamp technique to record the LFP from the pyramidal cell layer in rodent hippocampal slices, and reported that exogenous 5-HT decreased the power of kainate-induced gamma oscillations in mice and rats (Johnston et al., 2014).

However, under the specific pharmacological condition during esketamine anesthesia, *Wang and colleagues* observed a positive correlation between 5-HT concentration in the mPFC and esketamine-induced enhancement of gamma oscillations (Wang et al., 2022), suggesting that the inhibitory role of 5-HT might be altered or overridden by the drug's primary action on NMDA receptors. In summary, this series of findings solidifies the role of 5-HT as a predominantly inhibitory regulator of gamma oscillations, with implications for understanding the neurophysiological mechanisms underlying mood disorders and their treatments.

7.2 5-HT_{1A} receptor

The 5-HT_{1A} receptor (5-HT_{1A}R), as an important subtype among 5-HT receptors, plays a key role in the development of depression (Kaufman et al., 2016) and has garnered significant research interest for its role in regulating gamma oscillations. Emerging evidence suggested that it was the 5-HT_{1A}R that mediates 5-HT's inhibitory effects on gamma oscillations. For instance, building on the work by *Krause and Jia*, they observed that the selective 5-HT_{1A}R agonists 8-OH-DPAT and buspirone both exerted inhibitory effects on gamma oscillations, and the selective 5-HT_{1A}R antagonist WAY-100635 could block 5-HT's inhibitory effects on gamma oscillations (Krause & Jia, 2005). Further supporting this mechanism, *Johnston et al.* found that 5-HT receptor activation significantly decreases kainate-induced hippocampal gamma oscillations in mice and rats specifically via 5-HT_{1A}R (Johnston et al., 2014). Interestingly, *Xu and colleagues* demonstrated that although

5-HT_{1A}R activation was shown to attenuate theta-fast gamma cross-frequency coupling in the hippocampal CA1, it enhanced the ventral CA1-mPFC coupling in the gamma band (Xu et al., 2016). Regarding the mechanism underlying the mediating role of 5-HT_{1A}R in 5-HT's effects on gamma oscillations, it has been proposed that 5-HT might regulate gamma oscillations through 5-HT_{1A}R-expressing fast-spiking interneurons (FSi) (Puig et al., 2010). Given that FSi are pivotal in generating gamma oscillations, this suggests a plausible cellular mechanism through which 5-HTergic signaling could modulate network dynamics. Together, these findings contribute to a better understanding of how impaired serotonergic modulation of network activity may underlie psychiatric disorders such as depression.

7.3 Other 5-HT receptors

Beyond the 5-HT_{1A}R, emerging evidence indicates that several other 5-HT receptor subtypes are critically involved in modulating gamma oscillations, and these receptors mediate their effects through distinct, and sometimes interactive, cellular mechanisms. For instance, *Puig et al.* previously found that 5-HT_{2A}R regulated gamma oscillations, building on their observations that 5-HT might regulate gamma oscillations through 5-HT_{2A}R-expressing fast-spiking interneurons (FSi) (Puig et al., 2010). This role is further supported by studies on serotonergic hallucinogens, which primarily act through 5-HT_{2A}R (Halberstadt, 2015). For instance, *Goda et al.* demonstrated that serotonergic hallucinogen-induced changes

in the NAc gamma oscillation of freely moving rats are mediated, at least in part, by stimulation of 5-HT_{2A}R (Goda et al., 2013). Consistently, **Gener et al.** also reported that 5-HT_{2A}R antagonism with M100907 suppressed theta/gamma power in the PFC and HPC, and activation of 5-HT_{2A}R with DOI enhanced high-gamma oscillations in the PFC and PFC-HPC high-gamma functional connectivity, an effect potentially underlying its hallucinogenic properties (Gener et al., 2019). In addition, some studies highlight a significant role of the 5-HT₃R in regulating gamma oscillations, often through interactions with other neurotransmitter systems. For instance, 5-HT₃R might mediate BDNF's regulatory effects on hippocampal gamma oscillations (Huang & Morozov, 2011), indicating interactions between 5-HT₃R and BDNF in regulating gamma oscillations. Building on this work, **Huang & Morozov** found that 5-HT₃R blockade reverses hippocampal gamma oscillation deficits in BDNF knockout mice, and reduction of gamma oscillation power is partially reversed by 5-HT₃R antagonist tropisetron (30 nM). Furthermore, the modulation of gamma oscillations by 5-HT₃R appears to require functional crosstalk with the cholinergic system. **Skovgård and colleagues** found that the selective 5-HT₃R antagonist ondansetron (0.3 mg/kg, i.v.) alone did not affect hippocampal network oscillations; it potently enhanced and prolonged the gamma oscillations induced by the cholinesterase inhibitor donepezil (0.2-0.3 mg/kg) (Skovgård et al., 2018). Supporting a broader role, **Castañé et al.** found that FPPQ (a dual 5-HT₃/5-HT₆ receptor antagonist), endowed with pro-cognitive properties, attenuated

phencyclidine-induced hyperlocomotion and augmentation of gamma oscillations in the mPFC, dorsal hippocampus, and NAc of freely moving rats (Castañé et al., 2022). The mechanism underlying these effects appears to involve inhibitory interneuron regulation. For example, *Huang et al.* reported that 5-HT_{3A}R activation with its agonist *m*-CPBG impairs hippocampal carbachol-induced gamma oscillations and reduces inhibitory inputs to PV⁺ interneurons (Huang et al., 2016), indicating that 5-HT_{3A}R modulates hippocampal gamma oscillations by regulating synchrony of PV⁺ interneurons and highlighting an inhibitory pathway among interneuron subtypes as a key mechanism in the control of network oscillations.

7.4 Concluding remarks

In summary, converging evidence from pharmacological, genetic, and electrophysiological studies establishes 5-HT as a crucial regulator of gamma oscillations, predominantly through 5-HT_{1A}R-mediated inhibitory control. In addition, other subtypes (e.g., 5-HT_{2A}R, 5-HT_{3R}) contribute to context-dependent modulation, often acting through PV⁺ interneurons, which serve as a central cellular node for network synchronization. Nevertheless, some important questions remain, including reconciling 5-HT's inhibitory role with state-dependent facilitatory effects and elucidating how different receptor subtypes integrate at the microcircuit level. Future studies combining cell-type-specific manipulations and behaviorally-relevant *in vivo* recordings are essential to understand how adaptive serotonergic tuning of gamma oscillations supports mood regulation, and how this mechanism becomes

dysregulated in depression.

8 Gamma oscillations and dopamine system

Dopamine (DA), a catecholamine neurotransmitter that plays an important role in depression and antidepressants (Dailly et al., 2004; Malhi & Berk, 2007; Nestler & Carlezon, 2006), contributed a lot to regulating gamma oscillations according to recent reports (Andino-Pavlovsky et al., 2017; Wang et al., 2019) (**Table 4**). Accordingly, this section summarizes the interplay between gamma oscillations and the brain DA system, with a focus on the DA neurotransmitter itself and its specific receptor subtypes implicated in this regulation.

8.1 DA neurotransmitter

An emerging consensus prioritizes long-term dopaminergic adaptations over acute pharmacological effects in regulating gamma oscillations. Genetic mouse models have been utilized to investigate the effects of altered DA transmission on gamma oscillations. For instance, *Dzirasa's team* reported that DA transporter knock-out (DAT-KO) mice showed increased hippocampal (but not cortical) gamma oscillations, and enhanced gamma phase synchrony between the hippocampus and PFC during novelty-induced exploration (Dzirasa et al., 2009; Dzirasa et al., 2006). This finding is consistent with human evidence from *Demiralp's group* showing that carriers of polymorphisms in the DA transporter gene DAT1 that prolong event-related increases in DA show an increase in evoked gamma oscillations in response to target stimuli (Demiralp et al., 2007). However, another two groups demonstrated

that acute elevations of DA levels with pharmacological treatments, including amphetamine, apomorphine, or methamphetamine, had no significant effect on baseline or auditory-evoked gamma in the rodent cortex and hippocampus (Ehrlichman et al., 2009; Ma & Leung, 2000). Collectively, these findings indicate that long-term adaptations in dopaminergic tone (rather than brief, acute increases) appear to be a more critical determinant of gamma oscillations.

8.2 D1-3 receptors

Many studies have shown that dopaminergic regulation of gamma oscillations might be dependent on DA receptor subtypes, including D1R, D2R, D3R, and D4R, though their roles are complex and context-dependent. For example, early work by *Weiss's group* provided initial insights, demonstrating that DA reversibly decreased the carbachol-induced integrated gamma oscillation power in the CA3 of rat hippocampal slices. This inhibitory effect on gamma oscillation activity was mimicked by the D1R agonist SKF-38393 and partially antagonized by the D1R antagonist SCH-23390 (Weiss et al., 2003), pointing to a significant role of D1R in regulating gamma oscillations. Building on this work, they found that the D2R agonist quinpirole failed to depress the gamma oscillations, and the D2R antagonist sulpiride did not prevent the suppressive dopamine effect, suggesting a limited involvement of D2R in this particular form of modulation. However, the regulatory picture of DA receptors is not solely inhibitory. A recent study from *Xie's team* suggested that DA strongly enhanced kainate-induced gamma oscillations in the

CA3 of rat hippocampal slices, which were largely blocked by D1R antagonist SCH23390, and partially blocked by D2/3R antagonist raclopride (Xie et al., 2021). Beyond D1R and D2R, **Gilbert and colleagues** recently reported that chronic treatment with pramipexole (a D3R agonist) increases cortical gamma oscillations as measured by *in vivo* EEG recordings in female BALB/c and C57BL/6J mice (Gilbert et al., 2024). This is further supported by a more recent study by **Wang et al.** showing that D3R activation with PD128907 (a selective D3R agonist) induced excessive gamma oscillations in cortico-striatal circuits, and D3R antagonism with PG01037 (a selective D3R antagonist) suppressed gamma oscillations, firmly establishing D3R as a modulator of network gamma synchrony (Wang et al., 2025).

8.3 D4 receptor

Among all its receptors, dopamine D4 receptors (D4R) exhibited the highest affinity for DA (Rondou et al., 2010), and substantial evidence specifically underscores their importance in gamma oscillation. Notably, **Andersson and colleagues** have demonstrated that PD168077 (a selective D4R agonist) significantly increased gamma oscillation power, and this effect was blocked by L-745,870 (a D4R antagonist) (Andersson et al., 2012). Mechanistically, they concluded that D4R activation affected fast-spiking interneuron synchronization and thereby increased gamma oscillation power via an NMDA receptor-dependent mechanism (Andersson et al., 2012). This pro-oscillatory effect of D4R activation is supported by work in aged mice from **Wang and colleagues**, where D4R activation

with PD168077 similarly enhanced hippocampal gamma oscillations in aged mice (Wang et al., 2022). Building on this work, *Wang and colleagues* found that L-745,870 and the NMDA receptor antagonist could both reduce PD168077-boosted gamma oscillation in aged mice. Moreover, *Ji et al.* reported that D4R activation by RO 10-5824 (a D4R agonist) could boost gamma oscillations in the mPFC and inhibit neuroinflammation and improve cognitive performance induced by repeated LPS injections (Ji et al., 2020). Collectively, these findings establish D4R activation as a potent and specific mechanism for regulating gamma oscillations, with potential therapeutic implications.

8.4 Integrated view of dopaminergic modulation of gamma oscillations

Collectively, research indicates that DA regulates gamma oscillations through mechanisms operating at multiple timescales and receptor-specific pathways. While acute pharmacological intervention shows limited effects on gamma oscillations, genetic models and chronic adaptations reveal that sustained dopaminergic tone critically shapes oscillatory networks. In addition, the DA receptor landscape reveals specialized functions. For example, D1 receptors mediate context-dependent bidirectional modulation, D3 receptors consistently enhance cortical-striatal gamma synchrony, and D4 receptors produce robust pro-oscillatory effects through NMDA-dependent interneuron synchronization. These findings establish DA as a multi-layered modulator of gamma oscillations, where receptor-specific signaling enables fine-tuned, state-dependent regulation.

9 Gamma oscillations in antidepressant therapies

This section synthesizes evidence on how diverse antidepressant interventions, including pharmacological agents and neuromodulation therapies, modulate gamma oscillations to determine whether altering gamma activity is a common therapeutic pathway or a target for personalized intervention, aiming to inform the development of more precise therapies.

9.1 Highlights on SSRIs

As first-line pharmacological treatments, SSRIs have been frequently linked to gamma oscillations. However, the specific effects of SSRIs on gamma oscillations remain inconsistent across studies (**Table 5**). For instance, one group using extracellular LFP recordings and multiple unit activity (MUA) reported that acute administration (60 min) of citalopram (10-20 mg/kg, i.p.) significantly reduces the power and frequency of the early gamma oscillations in the barrel cortex of neonatal rats, while also suppressing the amplitude and prolongs the delay of the sensory evoked potentials (Akhmetshina et al., 2016). Similarly, *Papp et al.* reported that acute administration of escitalopram (10 mg/kg, i.p.), which is the (S)-enantiomer of the racemic citalopram and is effective and well tolerated in the treatment of depression in clinical studies (Yin et al., 2023), suppresses gamma oscillations during rapid eye movement sleep (REMS) and intermediate sleep, but chronic escitalopram treatment (10 mg/kg, osmotic minipumps) increased gamma oscillations during non-REMS stages (Papp et al., 2018). These findings suggested

that acute and chronic escitalopram affected gamma oscillations differently.

Recent EEG studies also indicated a connection between gamma synchronization and treatment response to paroxetine. For instance, *Arikan et al.* found that gamma oscillations in the frontal and central regions were significantly correlated with Hamilton Depression Rating Scale (HDRS) scores following a 6-week paroxetine treatment in patients during an episode of major depression (Arikan et al., 2018). More recent clinical EEG evidence from *Eskikurt et al.* further confirmed that 6-week paroxetine treatment (4 mg/kg) led to an increase in gamma oscillations (Eskikurt et al., 2024). Nonetheless, further investigation is warranted to fully elucidate paroxetine's impact on neural oscillations.

The available evidence also points to fluoxetine's modulatory role in gamma oscillations. Earlier *in vitro* work from *Méndez et al.* demonstrated that fluoxetine (20 μ M) significantly suppressed kainate-induced gamma oscillations in hippocampal CA1 slices even after preincubation with amine transporter inhibitors, indicating this effect is independent of monoaminergic actions (Méndez et al., 2012). Our recent *in vivo* study suggested that 21-day fluoxetine treatment (10 mg/kg, i.g.) significantly increased gamma oscillations in the mPFC of mice and elicited antidepressant-like effects, whereas acute (60 min) or sub-chronic fluoxetine treatment (once daily for 7 days) had no such effect (Yin et al., 2024). In addition, *Atiyah's team* recently revealed that acute administration of fluoxetine (20 mg/kg, i.g.) enhanced high-frequency gamma oscillations in the NAc and dorsal

hippocampus (dHP) during both mobility and immobility states, while attenuating stress-induced reductions in gamma power observed in the control group (Atiyah et al., 2025). Collectively, a deeper understanding of these electrophysiological properties is essential for addressing the heterogeneous treatment responses to antidepressants in clinical practice.

9.2 Highlights on other antidepressants

Research on the effects of tricyclic antidepressants (TCAs) and SNRIs on gamma oscillations remains relatively limited (**Table 5**). *Méndez et al.* found that extracellular LFP recordings from stratum radiatum in hippocampal CA1 showed that imipramine (20 μ M), a classical TCA, strongly altered GABA release from synapses formed by fast-spiking (FS) cells in rodents and reduced gamma oscillation power (Méndez et al., 2012), indicating that strong alteration of FS cell-mediated alteration of perisomatic inhibition by imipramine could generate an acute disruption of gamma oscillations. Moreover, *Alaiyed et al.* found that 2-week administration of venlafaxine (30 mg/kg, i.p.), a classical SNRI, increased carbachol-induced gamma oscillations in ex vivo hippocampal slices of mice (Alaiyed et al., 2019), accompanied by increases in hippocampal matrix metalloproteinase (MMP)-9 levels, suggesting that MMP-9 contributes to venlafaxine-enhanced gamma oscillations. In addition, *Leiser and colleagues* reported that vortioxetine, a new serotonergic antidepressant (D'Agostino et al., 2015), significantly increased gamma oscillation power in actively awake rats (Leiser et al., 2014). Notably, an earlier study by *Hajós*

et al. showed that systemic administration of reboxetine, a selective noradrenaline reuptake inhibitor, synchronized hippocampal EEG and resulted in an increase in frequency and power of gamma-wave activity (Hajós et al., 2003). This finding highlights the potential importance of the noradrenergic system in the modulation of gamma oscillations.

9.3 Highlights on ketamine

A single infusion of subanesthetic-dose ketamine, a noncompetitive NMDA receptor antagonist, rapidly (within hours) relieves depressive symptoms in clinical and preclinical studies (Abbar et al., 2022; Yin et al., 2021; Zolghadriha et al., 2024), leading to the FDA approval of intranasal esketamine as a conjunctive treatment for TRD in 2019. Intriguingly, many studies highlight the important role of antidepressant-dose ketamine in modulating gamma oscillations (Gilbert & Zarate, 2020) (**Table 6**), providing new insights into developing next-generation rapid-onset antidepressants.

Preclinical evidence in rodents with paired ECoG and intracortical LFP recordings has revealed that acute administration of a low-dose ketamine significantly enhanced gamma oscillations in intracortical networks within 40 min (Hakami et al., 2009). More recently, *Kantor et al.* recorded the EEG signals in the Wistar-Kyoto (WKY) rats, a suitable model of TRD, and reported that ketamine significantly increased gamma oscillations during wakefulness in both strains, but gamma oscillations were almost twice as large in Wistar-Kyoto (WKY) rats as in Sprague-Dawley (SD) rats

(Kantor et al., 2023). Supporting these findings, our recent study revealed that a low-dose ketamine (10 mg/kg, i.p.) treatment for 24 h significantly increased gamma oscillations in the mPFC of mice (Yin et al., 2024). Importantly, further findings in non-human primates from *Slovik and colleagues* that acute ketamine (10 mg/kg, i.m.) infusion increased gamma oscillations in the primary motor cortex (M1) and the external globus pallidus (GPe) of African Green monkeys (Slovik et al., 2017), reinforced ketamine's effects on gamma oscillations. Collectively, these preclinical studies across different rodent and non-human primate models establish that ketamine robustly enhances gamma oscillations, an effect that is both rapid and potentially modulated by underlying neurobiology, underscoring its translational relevance as a key electrophysiological signature of ketamine's action.

Clinical studies have similarly reported ketamine-induced enhancements in gamma oscillations. For instance, a MEG study from *Muthukumaraswamy's team* reported that compared with placebo sessions, subanesthetic-dose ketamine significantly increased the high-gamma oscillations in the parietal and cingulate cortex of healthy subjects (Muthukumaraswamy et al., 2015). This finding was corroborated by a subsequent single-blind crossover MEG study from the same team, which reported that ketamine (0.5 mg/kg, i.v.) amplifies gamma-band amplitudes in the motor and visual cortices (Shaw et al., 2015). Further support comes from EEG studies documenting increased resting-state gamma activities in healthy individuals following ketamine administration (de la Salle et al., 2016). In addition, a

randomized and placebo-controlled crossover simultaneous fMRI/EEG study pointed to a shift of energy to gamma wave frequencies was seen in the subanesthetic ketamine condition (Zacharias et al., 2020). Importantly, the electrophysiological response to ketamine differs between healthy subjects and depressed patients, and even among patients based on clinical outcome. For instance, *Cornwell and colleagues* demonstrated that only ketamine responders uniformly exhibited increased gamma oscillations post-ketamine infusion compared to baseline (Cornwell et al., 2012), whereas non-responders demonstrated no such change. More importantly, the baseline neural state appears to be a critical moderator of treatment efficacy. In a double-blind, placebo-controlled, randomized crossover MEG study, *Nugent et al.* revealed that among depressed patients showing a large increase in gamma oscillation power after ketamine (0.5 mg/kg), those with lower baseline gamma levels had a superior antidepressant response compared to those with higher baseline levels (Nugent et al., 2019).

Collectively, evidence from both rodent and human studies indicates that the potentiation of gamma oscillations is a robust electrophysiological response to ketamine, positioning it as a promising biomarker for predicting antidepressant efficacy.

9.4 Highlights on neuromodulation therapies

Gamma oscillations, including resting-state EEG gamma power and theta-gamma coupling, have been strongly linked to depression and are modifiable through

various neuromodulation interventions (**Figure 2**). Both rTMS and DBS have emerged as effective treatments for depression (Kinjo et al., 2021; Pathak et al., 2016; Scangos et al., 2021), with gamma oscillations serving as key indicators of treatment response.

Several rTMS studies support the role of gamma oscillations in therapeutic outcomes. For example, *Pathak and colleagues* reported that patients with depression displayed increased gamma oscillations at the left-dorsolateral prefrontal cortex (L-DLPFC) after 4-week rTMS treatment, which was correlated with improvement in depression symptoms (Pathak et al., 2016). Similarly, *Noda et al.* observed that increased resting gamma oscillations in depressed patients following rTMS were significantly correlated with improvements in HAM-D17 and BDI scores (Noda et al., 2017), suggesting its potential as a biomarker of depression associated with the therapeutic effects of rTMS. Supporting this, *Bailey and colleagues* found that responders to rTMS for patients with TRD showed an increase in gamma connectivity from baseline to week 1, with a concurrent improvement in mood and reaction times of the working memory task (Bailey et al., 2018). However, not all findings align uniformly. *Kazemi and colleagues* noted that rTMS-induced recovery in bipolar depression was associated with a decrease in gamma oscillations in the frontal and cingulate regions (Kazemi et al., 2016), highlighting diagnostic and methodological nuances.

DBS has shown promise in modulating gamma oscillations in depression. For

example, *Sun et al.* reported that DBS in patients with TRD suppressed gamma oscillations during cognitive tasks (N-back), correlating with symptom reduction (Sun et al., 2015). Similarly, *Scangos and colleagues* implemented a biomarker-driven closed-loop therapy in an individual with depression with chronic DBS and found that it resulted in a rapid and sustained improvement in clinical symptoms, accompanied by alterations in gamma oscillations (Scangos et al., 2021). Furthermore, *Scherer et al.* found that responders to the subcallosal cingulate DBS (SCC-DBS) for TRD showed decreased gamma oscillations within nodes of the default mode, central executive, and somatomotor networks, Broca's area, and lingual gyrus (Scherer et al., 2023). Furthermore, preclinical studies provide mechanistic insights. *Akhoondian et al.* showed that DBS in the lateral habenula (LHb) alleviated CMS-induced depression-like behaviors and reversed CMS-induced increase in the gamma bands in the NAc (Akhoondian et al., 2023). However, *Zhou et al.* reported that DBS in the NAc rescued CUMS-induced depression-like behaviors and reversed high gamma oscillation reduction and neurogenesis impairment in the dorsal dentate gyrus (Zhou et al., 2022). Taken together, DBS outcomes appear more variable, possibly depending on stimulation targets and baseline pathology. Further research is needed to clarify these dynamics and optimize gamma-based treatment personalization.

9.5 40 Hz flickering light therapy

Importantly, the therapeutic potential of sensory-evoked gamma oscillations,

particularly through 40 Hz light flicker, in treating depression is increasingly recognized, with evidence pointing to multi-level neuro-modulatory and circuit-remodeling effects. Importantly, 40 Hz induced gamma enrichment within 15 min after light flickers (Tian et al., 2021), and 40 Hz light flicker significantly affected many brain regions that are related to chronic stress. For example, **Lin**'s team found that 40 Hz blue light changes hippocampal activation and functional connectivity underlying recognition memory (Lin et al., 2021). At the molecular and cellular level, **our team** recently found that rhythmic 40 Hz light stimulation mitigates stress-induced dendritic spine alterations, upregulates hippocampal and prefrontal expression of GluN2A and GluA1, and reduces microglial activation along with M1 phenotype markers such as iNOS and CD68, thereby alleviating stress-related behavioral and cognitive deficits (Yao et al., 2024). Furthermore, in a recent pre-clinical study, **Barzegar Behrooz** and his colleagues found that 40 Hz flickering light for 15 min daily over 7 days significantly improved rats' preferences on the Morris water maze test and novel object recognition test, as well as long-term potentiation, indicating its critical role in preserving cognitive function and synaptic plasticity (Barzegar Behrooz et al., 2024). Collectively, these findings establish 40 Hz light flicker as a promising intervention for rescuing molecular impairments and restoring neural synchrony, highlighting its translational potential for depression treatment.

10 Future perspectives and conclusions

Investigations into gamma oscillations have shifted the field of depression

research from static chemical models toward a dynamic, systems-level understanding. Converging evidence now underscores that gamma oscillation abnormalities represent a core pathophysiological feature of depression, linking molecular deficits to large-scale network dysfunction. As this review has synthesized, gamma oscillations are not only perturbed in the pathophysiology of depression but are also critically modulated by a range of antidepressant interventions, including monoaminergic drugs, ketamine, and neuromodulation therapies. These rhythms sit at the functional intersection of key neurotransmitter systems and synaptic plasticity, positioning them as a central integrator of brain states relevant to depression. This paradigm opens promising translational avenues. For example, quantifying gamma oscillations might provide an objective biomarker for diagnosis, prediction of treatment response, and identification of patient subtypes. Together, these advances position gamma oscillations as both a biomarker and a therapeutic target, paving the way for more precise and effective management of depression.

In conclusion, focusing on gamma oscillations provides a unifying framework that bridges molecular alterations, circuit-level dysfunction, and clinical manifestations in depression. Although challenges such as measurement standardization and causal inference remain, further elucidating the role of gamma oscillations holds considerable potential to reshape diagnostic strategies and inspire novel therapeutics for this debilitating disorder.

Author contributions

All authors read and approved the final manuscript. **Yong-Yu Yin** wrote the original draft and prepared the **Figures and Tables** used in this manuscript. **Yun-Feng Li** reviewed and edited this manuscript.

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Data Availability

No data was generated or analyzed during this review.

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Figure legends

Figure 1. Neural oscillation waves based on EEG recordings. Neural oscillations are classified based on frequency into the delta (0.1-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-30 Hz), and gamma (30-80 Hz) oscillations.

Figure 2. Neuromodulation therapies for depression, such as rTMS and DBS.

Table legends

Table 1. Summary of relationships between gamma oscillations with depression in preclinical studies.

Table 2. Summary of relationships between gamma oscillations with depression in clinical studies.

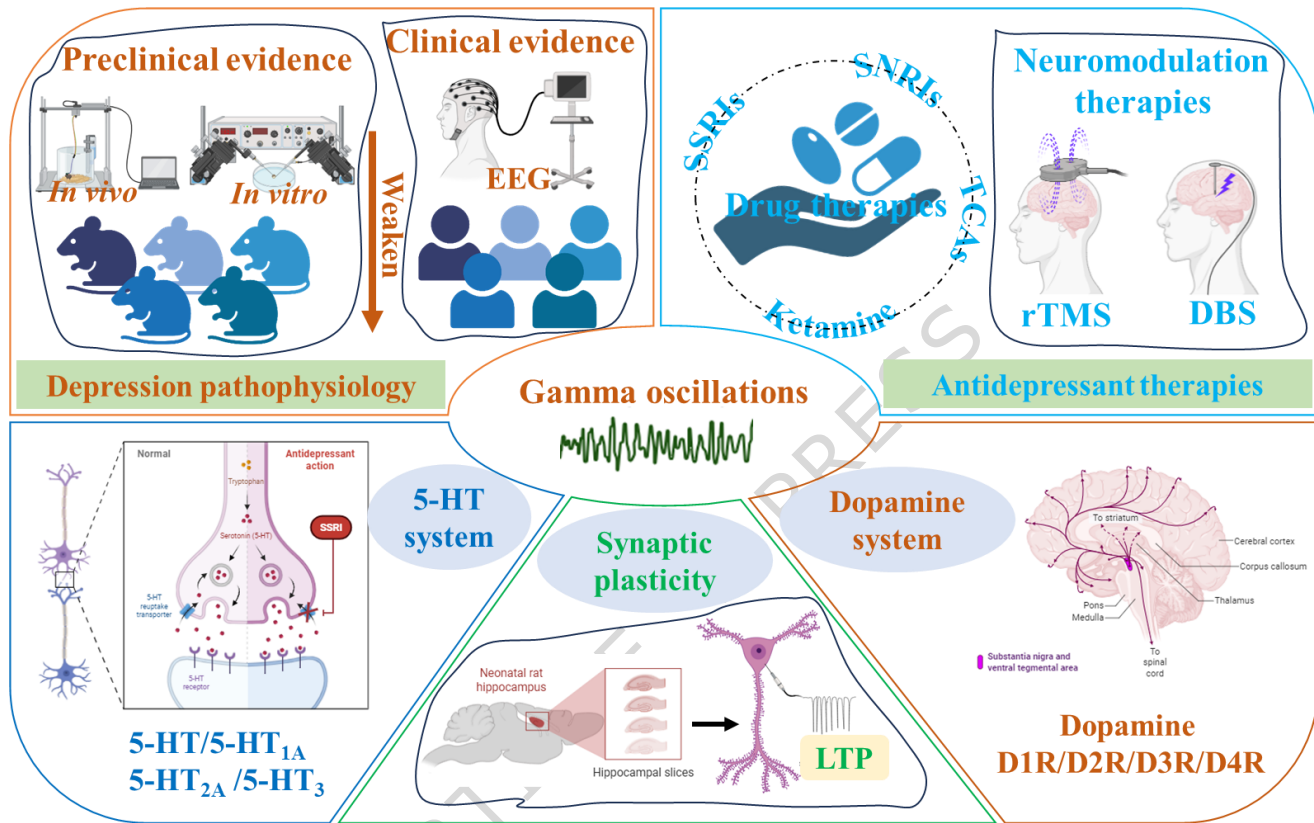
Table 3. Summary of relationships between gamma oscillations with 5-HT system.

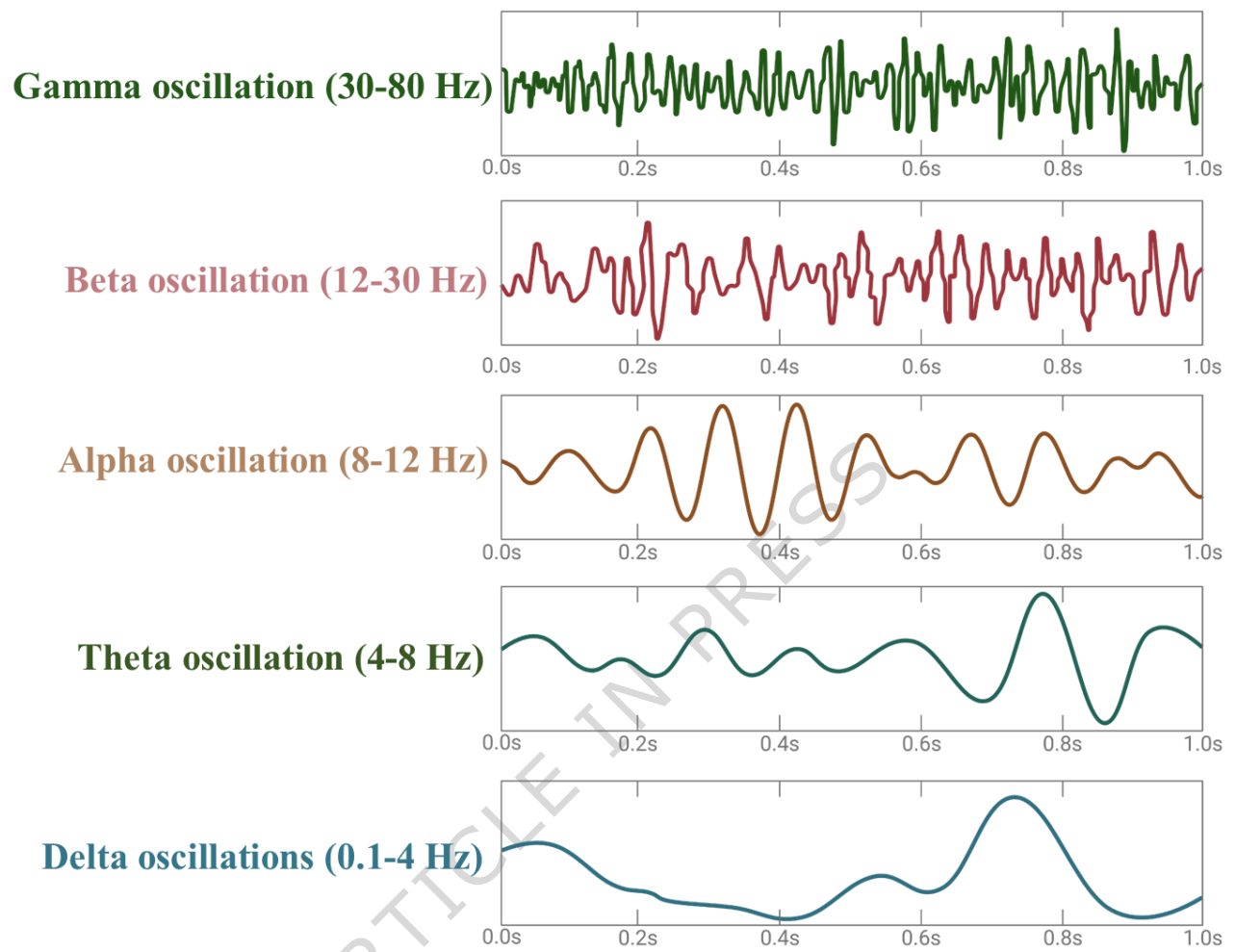
Table 4. Summary of relationships between gamma oscillations with dopamine system.

Table 5. Summary of relationships between gamma oscillations with antidepressants.

Table 6. Summary of relationships between gamma oscillations with ketamine in

preclinical and clinical studies.





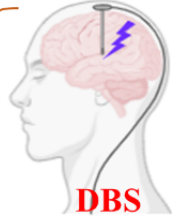
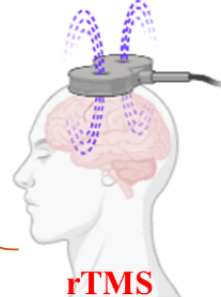
	Methods	Target(s)	Current	Invasive
Neuromodulation therapies for depression	 DBS	◆ Subcallosal cingulate ◆ Lateral habenula ◆ Nucleus accumbens	◆ Low Frequency 8-20 Hz ◆ High Frequency 40-250 Hz ◆ 1- 20 Voltages	Yes
	 rTMS	◆ Right/left-dorsolateral prefrontal cortex	◆ Single or repeated sessions ◆ Low Frequency 5-20 Hz	No

Table 1. Summary of relationships between gamma oscillations with depression in preclinical studies.

Subjects	Methods	Model	Main findings	References
Disc1 mice > 5 weeks	<i>In vivo</i> <i>LFPs</i>	No	Disc1 mice displayed depression-related behavioral despair in the SPT, TST, and FST; these behavioral deficits were accompanied by a marked reduction in gamma oscillations in the prelimbic cortex (PrLC).	(Sauer et al., 2015)
FSL rats	<i>In vivo</i> <i>LFPs</i>	No	FSL rats exhibited reduced alpha, beta, and low-gamma oscillatory activity in the mPFC and NAc, whereas the subthalamic nucleus showed decreased alpha and beta but increased low-gamma activity.	(Voget et al., 2015)
C57BL/6 25-30 g	<i>In vivo</i> <i>LFPs</i>	CRS 4 weeks	CRS significantly induced depression-like behaviors and decreased the firing of pyramidal neurons and the activities of gamma oscillations in the mPFC. There are negative correlations between CRS-induced immobility durations in the FST between low- and high-gamma oscillations in the mPFC.	(Yin et al., 2023 ; Yin et al., 2024)
C57BL/6J mice PV-Cre mice	<i>In vivo</i> <i>LFPs</i>	CUMS 2 weeks	CUMS induced depression-like behaviors, reduced high gamma oscillation, and induced neurogenesis impairment in the dorsal dentate gyrus of the hippocampus, which were rescued by NAc-DBS.	(Zhou et al., 2022)
Wistar rats 250-300 g	<i>In vivo</i> <i>LFPs</i>	CMS 4 weeks	CMS enhanced nearly all oscillatory bands in the NAc and increased NAc-PrL coherence in low frequencies. Opposing these effects, LHb-DBS	(Akhoondian et al., 2023)

			reduced power across multiple bands (NAc theta-alpha-beta-gamma; PrL low-gamma) while selectively enhancing NAc-PrL coherence in the beta and low-gamma range.	
C57BL/6 mice 3 months; Long-Evens rats 3-4 months	<i>In vivo</i> <i>LFPs</i>	OBx Bilateral	OBx dramatically reduced gamma oscillations, notably in the piriform cortex (PirC), which was accompanied one month later by depression-like behaviors, including anxiety and anhedonia. This supports the conclusion that the olfactory bulb drives brain-wide coherent gamma oscillations essential for a healthy mood.	(Li et al., 2023)
C57BL/6 25-30 g	<i>In vivo</i> <i>LFPs</i>	LPS 0.5 mg/kg, i.p.	A single LPS injection remarkably decreased low- and high-gamma oscillations in the mPFC, accompanied by significant depression-like behaviors in mice, and there are negative correlations between LPS-induced immobility durations between low- and high-gamma oscillations in the mPFC.	(Yin et al., 2024)
C57BL/6 mice 3-4 months	<i>In vivo</i> <i>LFPs</i>	LPS, 3mg/kg, i.p. CUS, > 5 weeks	Mice exposed to LPS and CUS exhibited significantly reduced PV expression and disturbed theta/gamma oscillations in the PFC, alongside behavioral deficits including anhedonia, cognitive impairment, and anxiety-like symptoms.	(Liu et al., 2023)
C57BL/6 6-8 weeks	<i>In vivo</i> <i>LFPs</i>	LPS 1 mg/kg, i.p.	Repeated LPS injections induced dysfunction of PV ⁺ interneurons and reduced gamma oscillations in the hippocampus; conversely, neuregulin-1 treatment reversed these LPS-induced deficits in both PV expression and	(Gao et al., 2023)

gamma oscillations.

Table 2. Summary of relationships between gamma oscillations with depression in clinical studies.

Subjects	State	Methods	Main findings	Reference
MDD patients with first episode	Resting Tasks	<i>In vivo</i> <i>EEG</i>	Gamma rhythm synchronization between cortical areas remained unchanged from rest to cognitive loading in healthy subjects, but was elevated during stress and in patients with depression.	(Strelets et al., 2007)
Depression patients	Tasks	<i>In vivo</i> <i>EEG</i>	Patients demonstrated elevated gamma oscillations. In contrast, healthy controls exhibited higher global gamma coherence in response to negative conditions compared to positive conditions.	(Li et al., 2015)
MDD patients	Resting	<i>In vivo</i> <i>EEG</i>	Significantly increased complexity was observed not only in full-band activity but also in gamma sub-bands of the EEG.	(Akar et al., 2015)
MDD patients with suicidal-ideation	Resting	<i>In vivo</i> <i>qEEG</i>	Individuals with suicide ideation exhibited elevated high-gamma oscillations in several regions relative to other groups, providing preliminary evidence that it may serve as a biomarker to identify suicidality.	(Arikan et al., 2019)
MDD patients with suicidal-ideation	Tasks	<i>In vivo</i> <i>MEG</i>	Gamma oscillations were significantly enhanced in patients, primarily localized within regions of the fronto-parietal control network (FPN), visual network, default mode network (DMN), and salience network. Furthermore, inter-network gamma connectivity between the FPN and DMN	(Dai et al., 2022)

			showed a significant negative correlation with suicide risk, which remained unaffected by depression severity.	
TRD patients	Resting	<i>In vivo</i> <i>EEG</i>	Gamma oscillations in TRD patients demonstrated a positive correlation with scores on the 17-item Hamilton Depression Rating Scale (HDRS-17), suggesting their potential as an electrophysiological marker of depression severity. Conversely, the Sham group exhibited significantly stronger changes in alpha-beta amplitude modulation of gamma oscillations than both active rTMS groups.	(Tsai et al., 2025)
MDD/BD patients	Tasks	<i>In vivo</i> <i>MEG/MRI</i>	Gamma oscillations were significantly reduced in the frontal regions of both BD and MDD patients. In contrast, distinct oscillation patterns emerged in other areas: MDD patients exhibited increased gamma oscillations in the bilateral temporal regions, whereas BD patients showed an increase in the alpha-beta rhythm.	(Lee et al., 2010 ; Liu et al., 2014)
MDD patients with first-episode	Resting	<i>In vivo</i> <i>EEG</i>	Patients with MDD exhibited significantly reduced gamma oscillation power in the left temporal and bilateral occipital regions, as well as weakened gamma-band connectivity between the left hemisphere and the right frontal area.	(Liu et al., 2022)
MDD patients with suicide attempt	Tasks	<i>In vivo</i> <i>MEG</i>	Patients significantly exhibited attenuated activation in low gamma oscillations and reduced intra-network connectivity within the fronto-parietal region, while simultaneously displaying increased activation and hyper-connectivity between visual regions and the limbic system..	(Dai et al., 2023)

Table 3. Summary of relationships between gamma oscillations with 5-HT system.

5-HT system	Subjects	Methods	Main findings	Reference
5-HT 5-HT _{1A} R	SD rats 18-25 days	<i>In vitro</i> <i>Patch-clamp</i>	5-HT significantly decreases the power of carbachol-induced hippocampal gamma oscillations in a concentration-dependent and reversible manner. This inhibitory effect was mimicked by the 5-HT _{1A} R agonists 8-OH-DPAT and buspirone, and was blocked by the selective 5-HT _{1A} R antagonist WAY-100635, indicating the critical involvement of the 5-HT _{1A} R.	(Krause & Jia, 2005)
5-HT	Wistar rats 160-220 g	<i>In vitro</i> <i>Patch-clamp</i>	5-HT produced a dose-dependent and reversible suppression of kainate- and carbachol-induced gamma oscillations in the hippocampus. Notably, this suppression was mimicked by fenfluramine, consistent with its role as a compound that promotes presynaptic 5-HT release.	(Wójtowicz et al., 2009)
5-HT 5-HT _{1A} R 5-HT _{2A} R	Wistar rats 150-350 g	<i>In vivo</i> <i>LFPs</i>	Electrical stimulation-induced 5-HT significantly regulates prefrontal gamma oscillations through both the 5-HT _{1A} R and 5-HT _{2A} R, inducing an overall decrease in gamma oscillation amplitude. Notably, this effect arises from an underlying fine-tuning of the amplitude via 5-HT _{2A} R- and 5-HT _{1A} R-expressing fast-spiking interneurons (FSi).	(Puig et al., 2010)

5-HT 5-HT _{1A} R	SD rats Postnatal day 15-30; PV- GFP mice Postnatal day 14-27;	<i>In vitro</i> <i>Patch-clamp</i>	Whole-cell patch clamp recordings demonstrated that 5-HT decreases the power of kainate-induced hippocampal gamma oscillations in rats and mice via the 5-HT _{1A} R subtype, through hyperpolarizing CA3 pyramidal cells and reducing their firing frequency.	(Johnston et al., 2014)
5-HT	C57BL/6J mice	<i>In vivo</i> <i>EEG</i>	EEG recordings revealed a positive correlation between the esketamine-induced increase in gamma oscillation and the concentration of 5-HT in the mPFC.	(Wang et al., 2022)
5-HT _{2A} R	Wistar rats 250-350 g	<i>In vivo</i> <i>LFPs</i>	Serotonergic hallucinogens rapidly decreased the power of low-, but not high-gamma oscillations, and the 5HT _{2A} R antagonist MDL 11939 fully reversed DOI-induced changes on the low-gamma oscillations.	(Goda et al., 2013)
5-HT _{2A} R	C57BL/6 mice 2-3 months	<i>In vivo</i> <i>LFPs</i>	5-HT _{2A} R antagonism with M100907 reduced the locomotor activity and suppressed theta/gamma power in the PFC and HPC; in contrast, activation of 5-HT _{2A} R with DOI enhanced high-gamma oscillations in the PFC and PFC-HPC high-gamma functional connectivity, an effect potentially underlying its hallucinogenic properties.	(Gener et al., 2019)
5-HT ₃ R	BDNF KO mice 6-7 weeks	<i>In vitro</i> <i>Patch-clamp</i>	5-HT ₃ R blockade reverses hippocampal gamma oscillation deficits in BDNF knockout mice, and reduction of gamma oscillation power is partially reversed by 5-HT ₃ R antagonist tropisetron, indicating that 5-HT ₃ R might mediate BDNF's regulatory effects on gamma oscillations.	(Huang & Morozov, 2011)

5-HT ₃ R	SD rats 250-400 g	<i>In vivo</i> <i>LFPs</i>	The selective 5-HT ₃ R antagonist ondansetron alone did not affect hippocampal network oscillations. However, it potently enhanced and prolonged the theta and gamma oscillations induced by the cholinesterase inhibitor donepezil.	(Skovgård et al., 2018)
5-HT ₃ R 5-HT ₆ R	Wistar rats 250-300 g	<i>In vivo</i> <i>LFPs</i>	Endowed with pro-cognitive properties, the dual 5-HT ₃ R/5-HT ₆ R antagonist FPPQ suppressed phencyclidine-induced hyperlocomotion and the concomitant enhancement of gamma oscillations across the mPFC, dorsal hippocampus, and NAc in freely moving rats.	(Castañé et al., 2022)
5-HT _{3A} R	5-HT _{3A} R-KO mice 6-7 weeks	<i>In vitro</i> <i>Patch-clamp</i>	5-HT _{3A} R activation with 5-HT _{3A} R agonist <i>m</i> -CPBG impairs hippocampal carbachol-induced gamma oscillations in the hippocampal CA1 slices of wild-type mice; 5-HT _{3A} R activation by <i>m</i> -CPBG reduces inhibitory inputs to PV+ interneurons and desynchronizes their firings, indicating that 5-HT _{3A} R modulates hippocampal gamma oscillations by regulating synchrony of PV+ interneurons.	(Huang et al., 2016)

Table 4. Summary of the relationships between gamma oscillations with dopamine system.

Dopamine system	Subjects	Methods	Main findings	Reference
Dopamine	Wistar rats 2-3 months	<i>In vivo</i> <i>LFPs</i>	Dopamine release in the medial prefrontal cortex (mPFC) induces a shift in phase-amplitude comodulation from	(Andino-Pavlovsky et al., 2017)

			theta-gamma to delta-gamma.	
Dopamine	Reelin ^{+/-} mice 3-6 months	<i>In vitro</i> <i>Patch-clamp</i>	Dopamine increased hippocampal gamma oscillations in wild-type mice but not in heterozygous reeler mice.	(Wang et al., 2019)
Dopamine	DAT-KO mice NR1-KD mice	<i>In vivo</i> <i>LFPs</i>	DAT-KO mice showed increased hippocampal (but not cortical) gamma oscillations, and enhanced gamma phase synchrony between the hippocampus and PFC during novelty-induced exploration.	(Dzirasa et al., 2009 ; Dzirasa et al., 2006)
Dopamine	Human	<i>In vivo</i> <i>EEG</i>	The DAT1 polymorphism, which reduces dopamine transporter (DAT) expression and thereby increases extracellular dopamine, specifically potentiated evoked gamma oscillations in response to target stimuli.	(Demiralp et al., 2007)
Dopamine	C57BL/6Hsd mice 10-13 weeks	<i>In vivo</i> <i>EEG</i>	Acute elevations of DA levels with pharmacological treatments, like amphetamine, had no significant effect on baseline or auditory-evoked gamma in the rodent cortex and hippocampus.	(Ehrlichman et al., 2009)
D1R	Wistar rats 250 g	<i>In vitro</i> <i>Patch-clamp</i>	DA reversibly decreased the carbachol-induced integrated gamma oscillations power by 62% in the CA3 of rat hippocampal slices. This inhibitory effect on gamma oscillation activity was replicated by the D1R agonist SKF-38393 and partially blocked by the D1R antagonist SCH-23390.	(Weiss et al., 2003)
D1R/D2R/D3R	SD rats 4-5 weeks	<i>In vitro</i> <i>Patch-clamp</i>	Dopamine (DA) robustly enhanced kainate-induced gamma oscillations in the CA3 region of rat hippocampal slices. This enhancement was largely blocked by the D1R antagonist SCH23390 and partially suppressed by the D2/3R	(Xie et al., 2021)

antagonist raclopride.

D3R	BALB/c mice C57BL/6L mice 10-14 weeks	<i>In vivo</i> <i>EEG</i>	Chronic treatment with pramipexole (a D3R agonist) increases cortical gamma oscillations in female BALB/c and C57BL/6J mice.	(Gilbert et al., 2024)
D3R	Male rats 280-300 g	<i>In vivo</i> <i>LFPs</i>	D3R activation with PD128907 (a selective D3R agonist) induced excessive gamma oscillations in cortico-striatal circuits, and D3R antagonism with PG01037 (a selective D3R antagonist) suppressed gamma oscillations, firmly establishing D3R as a modulator of network gamma synchrony.	(Wang et al., 2025)
D4R	SD rats	<i>In vitro</i> <i>Patch-clamp</i>	The selective D4R agonist PD168077 significantly increased gamma oscillations in the hippocampal CA3, and this effect is blocked by the selective D4R antagonist L-745,870.	(Andersson et al., 2012)
D4R	C57BL/6J mice 22-28 months	<i>In vitro</i> <i>Patch-clamp</i>	Gamma oscillations in the hippocampal CA3 were significantly decreased in aged mice. While exogenous DA application restored these oscillations, this rescue was mimicked by the D4R agonist PD168077. Furthermore, the effect of PD168077 was blocked by the D4R antagonist L-745,870 and substantially attenuated by an NMDAR antagonist, indicating the involvement of both receptor pathways.	(Wang et al., 2022)
D4R	SD rats 10-12 weeks	<i>In vivo</i> <i>LFPs</i>	Gamma oscillations were positively associated with time spent in the novel object recognition test. Furthermore, D4R	(Ji et al., 2020)

activation with the selective D4R agonist RO-10-5824 could enhance gamma oscillations in the mPFC, inhibit neuroinflammation, and improve cognitive performance induced by repeated LPS injections.

Table 5. Summary of relationships between gamma oscillations with antidepressants.

Drugs	Subjects	Routine	Methods	Main findings	Reference
Citalopram	Wistar rats Postnatal day 2-29	10-20 mg/kg, i.p. Acute	<i>In vivo</i> <i>LFPs</i>	Citalopram significantly reduces the power and frequency of the early gamma oscillations in the barrel cortex of neonatal rats, with a concomitant suppression of the amplitude and prolongation of the latency of sensory evoked potentials.	(Akhmetshina et al., 2016)
Escitalopram	Wistar rats 250-280 g	10 mg/kg, i.p. Acute or 3 weeks	<i>In vivo</i> <i>EEG</i>	Acute escitalopram treatment suppresses gamma oscillations during rapid eye movement sleep (REMS) and intermediate sleep, but chronic escitalopram treatment increases gamma oscillations during non-REMS	(Papp et al., 2018)

				stages.	
Paroxetine	Human	6 weeks	<i>In vivo</i> <i>qEEG</i>	Following 6 weeks of paroxetine treatment, the strength of gamma oscillations in the frontal and central regions showed a significant correlation with the Hamilton Depression Rating Scale (HDRS) scores in patients with a major depressive episode.	(Arikan et al., 2018)
Paroxetine	Wistar rats 200-250 g	4 mg/kg, i.p. 6 weeks	<i>In vivo</i> <i>EEG</i>	Six-week paroxetine treatment led to an increase in gamma oscillations in healthy wakeful rats.	(Eskikurt et al., 2024)
Fluoxetine	Rodents Postnatal day 18-25	20 μ M	<i>In vitro</i> <i>Patch-clamp</i>	Fluoxetine significantly suppressed kainate-induced gamma oscillations in hippocampal CA1 slices even after preincubation with amine transporter inhibitors, indicating this effect is independent of monoaminergic actions.	(Méndez et al., 2012)
Fluoxetine	C57BL/6J mice 25-30 g	10 mg/kg, i.g. 3 weeks	<i>In vivo</i> <i>LFPs</i>	Only 21-day fluoxetine treatment (10 mg/kg, i.g.) significantly increased gamma oscillations in the mPFC and elicited antidepressant-like effects in mice, whereas	(Yin et al., 2024)

				neither acute (60 min) nor sub-chronic (7 days) regimens produced such effects.	
Fluoxetine	C57BL/6J mice 35-40 g	20 mg/kg, p.o. Acute	<i>In vivo</i> <i>LFPs</i>	Fluoxetine enhanced high-frequency gamma oscillations in the nucleus accumbens (NAc) and dorsal hippocampus (dHP) during both mobility and immobility states. It also counteracted the stress- induced suppression of gamma oscillations that was evident in the control group.	(Atiyah et al., 2025)
Imipramine	Rodents Postnatal day 18-25	20 μ M	<i>In vitro</i> <i>Patch-clamp</i>	Imipramine significantly suppressed kainate- induced gamma oscillations in hippocampal CA1 slices even after preincubation with amine transporter inhibitors, indicating this effect is independent of monoaminergic actions.	(Méndez et al., 2012)
Venlafaxine	MMP-9 null mice 4 weeks	30 mg/kg, i.p. 2 weeks	<i>In vitro</i> <i>LFPs</i>	Venlafaxine significantly increased carbachol-induced gamma oscillations in ex vivo hippocampal slices of mice. This effect was accompanied by a concomitant increase in hippocampal matrix	(Alaiyed et al., 2019)

metalloproteinase (MMP)-9 levels, pointing to a potential contributory role of MMP-9 in the venlafaxine-mediated enhancement of gamma oscillations.				
Vortioxetine	SD rats 250-300 g	1-10 mg/kg, i.p. Single	<i>In vivo</i> <i>EEG</i>	Vortioxetine, at doses of 5 and 10 mg/kg, increased gamma oscillation power in actively awake rats. This increase was also observed at doses of 1, 3, 5, and 10 mg/kg when vortioxetine was co-administered with flesinoxan. (Leiser et al., 2014)
Reboxetine	SD rats 270-300 g	0.3-1 mg/kg, i.v Acute	<i>In vivo</i> <i>EEG</i>	Reboxetine induced a significant synchronization of hippocampal EEG and increased the frequency and power of gamma-wave activity. (Hajós et al., 2003)

Table 6. Summary of relationships between gamma oscillations with ketamine in preclinical and clinical studies.

Subjects	Routes	Methods	Main findings	Reference
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Wistar rats 250-350 g	2.5-5 mg/kg, s.c. Acute	<i>In vivo</i> <i>ECoG/LFPs</i>	Acute ketamine injection produced a significant dose-dependent and immediate increase in both gamma oscillations and locomotor activity in freely moving rats. (Hakami et al., 2009)
Wistar-Kyoto rats 200-250 g	3-10 mg/kg, s.c. Acute	<i>In vivo</i> <i>EEG</i>	EEG recordings suggested that ketamine significantly increased the gamma oscillation power during wakefulness in both strains, but the gamma power was almost twice as large in WKY rats as in SD rats. (Kantor et al., 2023)
C57BL/6J Mice 25-30 g	10 mg/kg, i.p. Acute	<i>In vivo</i> <i>LFPs</i>	Ketamine treatment significantly increased low- and high-gamma oscillations in the mPFC, concurrent with the manifestation of significant antidepressant-like effects in mice. (Yin et al., 2024)
vervet Monkeys 3.3-4.4 kg	10 mg/kg, i.m. Acute	<i>In vivo</i> <i>LFPs</i>	Ketamine induced robust, spontaneous gamma oscillations in the primary motor cortex (M1) and the external globus pallidus (Slovik et al., 2017)

			(GPe), which appeared under the modulation of ultraslow oscillations and then evolved to exhibit strong synchronization both within and across these two brain regions.	
Human Healthy male subjects	0.375 mg/kg, i.v. Acute	<i>In vivo</i> <i>MEG</i>	Acute ketamine infusion increased high-gamma oscillations in the parietal and cingulate cortex of healthy subjects, suggesting that its antidepressant effects may be related to an enhanced regulation of frontoparietal connectivity.	(Muthukumaraswamy et al., 2015)
Human Healthy male subjects	0.5 mg/kg, i.v. Acute	<i>In vivo</i> <i>MEG</i>	An acute intravenous injection of ketamine significantly reduced the peak gamma frequency in the visual cortex. Concurrently, it significantly amplified gamma-band amplitudes in both the motor and visual cortices.	(Shaw et al., 2015)
Human Healthy male	0.65 mg/kg, i.v	<i>In vivo</i> <i>EEG</i>	Ketamine significantly increased the frequency-	(de la Salle et al., 2016)

subjects	Acute		specific current source density (CSD) within gamma frequencies, while significantly reducing the CSD in delta/theta and alpha frequencies.	
Human Healthy male subjects	0.1 mg/kg, i.v. Acute	<i>In vivo</i> <i>fMRI/EEG</i>	Ketamine induced a shift of spectral power toward delta, theta, and gamma frequency bands. Concurrently, it decreased functional connectivity in the mPFC. Notably, this frontal connectivity was negatively correlated with the power of both gamma and theta oscillations.	(Zacharias et al., 2020)
Human Patients with MDD	0.5 mg/kg, i.v. Acute	<i>In vivo</i> <i>MEG</i>	An increase in stimulus-evoked somatosensory cortical responses post-ketamine was observed only in treatment responders, not in non-responders.	(Cornwell et al., 2012)
Human Patients with MDD	0.5 mg/kg, i.v. Acute	<i>In vivo</i> <i>MEG</i>	Ketamine increased resting gamma oscillations in both unmedicated subjects with MDD and healthy subjects, and	(Nugent et al., 2019)

demonstrated that basal gamma oscillations serve as a moderator of relationships between post-ketamine gamma oscillations and antidepressant effects.

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